



Mai Nhat Duy



mainhatduy



mainhatduy



myduy



maiduy07102003@gmail.com



+84.944.043.973



mainhatduy.github.io

PROFILE

AI Engineer at **QUAVEO** and research assistant at **TDTU NLP-KD Lab** with a passion for exploring research and technologies in **Artificial Intelligence (AI)**, particularly interested in applying **Natural Language Processing (NLP)** to real-world problems. Experienced in working with **Google Cloud Platform (GCP)** and **Agent Development Kit (ADK)**, open-minded, proactive in learning, and knowledgeable about processes and techniques in training **Large Language Models (LLMs)**. Growth mindset, lifelong learner.

EDUCATION

Final-year student – Computer Science, **Ton Duc Thang University**

AWARDS

Encouragement Award – Southern Vietnam Student Artificial Intelligence Olympiad 2025

SKILLS

Model Training Frameworks: PyTorch, Unsloth, Transformer, DeepSpeed.
Programming Languages: Python, Java.
Other Tools: Git, ElasticSearch, LangGraph, LangChain, GCP, ADK.
Additional: English - Aptis ESOL – 152 points (CEFR Level B2)

PROJECTS

Enhancing AMR Parsing in LLMs via Reasoning & RLVR

[Link to Model Collection](#)

Developed a 2-stage framework combining **Chain-of-Thought (CoT)** data synthesis, **Supervised Fine-Tuning (SFT)**, and **Reinforcement Learning with Verifiable Rewards (RLVR)** using **GRPO** and **DAPO** to improve graph reasoning in LLMs. Designed a dual reward function (Format & Accuracy), achieving a SOTA **82.88% Smatch F1** score on the AMR 3.0 benchmark, outperforming the SPRING baseline.

Vietnamese Medical Small Language Model

[Link to Model Collection](#)

Applied **Marker** for medical document digitization, **Datatrove** for data cleaning, and **Unsloth** for training language models on limited GPU infrastructure. The training process consists of two phases: (1) **Continual Pre-training** and (2) **Supervised Fine-Tuning**, aimed at creating an efficient **small language model** (1.7B-4B parameters) capable of understanding and answering Vietnamese medical questions.

Diffusion Language Model for Vietnamese NER

[Link to Model](#)

Fine-tuned **Dream 7B (Diffusion LM)** for Vietnamese NER task using manually annotated data from the **Nested NER on Vietnamese COVID-19** dataset. Use SFT technique to fine-tune a diffusion LM for the NER task.

Diffusion Speech Recognition

Link to Project

Applied **Diffusion Language Models (Diffusion LM)** to Automatic Speech Recognition (ASR) tasks, exploring non-autoregressive text generation for audio transcription and speech processing.

Agentic Chatbot

Link to Project

Applied AI Agent design frameworks such as **LangGraph**, **LangChain**, and **Elasticsearch** (vector base) to develop a chatbot integrated with Agentic RAG, capable of **querying both vector and SQL databases simultaneously** when needed, to enhance flexibility and accuracy in information retrieval.

REFERENCES

Assoc. Prof. Dr. Le Anh Cuong (Thesis Advisor)	TDTU	 leanhcuong@tdtu.edu.vn
M.Sc. Nguyen Thanh An	TDTU	 nguyenthahan@tdtu.edu.vn
Lecturer Nguyen Huu An	TDTU	 nguyenhuan@tdtu.edu.vn